Physical Therapy in Sport 23 (2017) 58–66

Contents lists available at ScienceDirect

Physical Therapy in Sport

journal homepage: www.elsevier.com/ptsp

Original Research

Distinct cut task strategy in Australian football players with a history of groin pain

Suzi Edwards [a, b, *], Hiram C. Brooke [b], Jill L. Cook [c]

[a] School of Environmental & Life Sciences, The University of Newcastle, Ourimbah, New South Wales, Australia
[b] School of Exercise Science, Sport & Health, Charles Sturt University, Bathurst, New South Wales, Australia
[c] Faculty of Health Sciences, La Trobe University, Bundoora, Victoria, Australia

ARTICLE INFO

Article history:
Received 5 April 2016
Received in revised form
14 July 2016
Accepted 20 July 2016

Keywords:
Landing mechanics
Athletic groin injury
Australian football
Biomechanics

ABSTRACT

Objectives: This study aimed to explore the differences in the magnitude of movement variability and strategies utilized during an unanticipated cut task between players with and without a history of groin pain.
Design: Cross-sectional design.
Setting: Biomechanics laboratory.
Particiants: Male Australian football players with (HISTORY; n = 7) or without (CONTROL; n = 10) a history of groin pain.
Outcome measures: Three-dimensional ground reaction forces (GRF) and kinematics were recorded during 10 successful trials of an unanticipated cut task, and isokinetic hip adduction and abduction strength. Between-group differences were determined using independent-samples *t*-tests and the coefficient of variation (CV).
Results: Key substantial between-group differences identified were that the HISTORY group displayed decreased knee flexion and hip internal rotation, increased knee internal rotation and T12-L1 right rotation, and higher GRFs during the cut task. They also utilized three invariant systems (ankle, knee and T12-L1 joints), while being connected by a segment (hip and L5-S1 joints) that displayed increased lumbopelvic movement during the cut task, and decreased adductor muscle strength.
Conclusion: This identifies the need for clinical management of the lower limb and thoracic segment to improve functional movement patterns in athletes with a history of a groin injury.
© 2016 Elsevier Ltd. All rights reserved.

1. Introduction

Groin pain in athletes (Weir et al., 2015a) are one the most severe injuries in professional male soccer players (Werner, Hagglund, Walden, & Ekstrand, 2009). They are the leading overuse injury with a high prevalence (6.7 injuries per club per season) and recurrence rate (11%) in professional male Australian football players (Orchard, Seward, & Orchard, 2014). The high recurrence rates may be attributed to limitations in diagnostic terminology, examination, accuracy and rehabilitation methods due to varied presentations of groin injuries, complex musculature and coexisting pathologies (Drew, Osmotherly, & Chiarelli, 2014; Weir et al., 2015a). Compounding these issues is an incomplete knowledge of risk factors associated with groin injury, which if identified, could allow better recovery and prevention strategies. Known risk factors for groin injury include a previous history of acute (Engebretsen, Myklebust, Holme, Engebretsen, & Bahr, 2010) or non-specified groin pain (Gabbe, Finch, Wajswelner, & Bennell, 2004; Thorborg, Rathleff, Petersen, Branci, & Hölmich, in press), weakness of the hip adductor muscle group (Engebretsen et al., 2010; Hrysomallis, 2009) and decreased hip internal rotation range of motion (Tak et al., 2015; Verrall et al., 2007).

Groin injury has been associated with sudden change-of-direction maneuvers (cut task), and rapid acceleration and deceleration in sports (Chaudhari, Jamison, McNally, Pan, & Schmitt, 2014; Gabbe et al., 2004). By exploring if cut task techniques differ in players with and without a history of groin pain, clinicians may be able to intervene and decrease the risk of groin injury (Scase, Cook, Makdissi, Gabbe, & Shuck, 2006). Aside from pre-

* Corresponding author. School of Environmental & Life Sciences, The University of Newcastle, 10 Chittaway Road, Ourimbah, New South Wales, 2522, Australia.
E-mail address: Suzi.Edwards@newcastle.edu.au (S. Edwards).

http://dx.doi.org/10.1016/j.ptsp.2016.07.005
1466-853X/© 2016 Elsevier Ltd. All rights reserved.

planned change-of-direction, the cut task is often performed in an unanticipated manner in response to an external stimulus such as avoiding a player or the unpredictable bounce of a ball (Besier, Lloyd, Ackland, & Cochrane, 2001a). An unanticipated cut task (Besier et al., 2001a) with a defensive opponent (McLean, Lipfert, & van den Bogert, 2004) places higher loading on the knee joint compared to a pre-planned cut with no defensive opponent. This may also expose the groin musculature to injury due to force transference via the kinetic chain and across the pubic rami either side of the pubic symphysis. Whilst the importance of adequate cut task strategies has been investigated in other sports (Scase et al., 2006), no research was located to help identify movement patterns that may predispose players to groin injury.

A lack of movement variability has been associated with injuries (Brown, Bowser, & Simpson, 2012; Seay, Van Emmerik, & Hamill, 2011), in different tasks and environmental conditions, and may reflect rigid, inflexible motor behaviors with limited adaptability. Movement variability, defined as the variations in motor performance over multiple repetitions of a task (James, Dufek, & Bates, 2000). Now researchers postulate that a greater level of movement variability may be a mechanism that protects tissue from overload and damage (Hamill, Palmer, & Van Emmerik, 2012; James et al., 2000) as it provides more flexibility to adapt to environmental conditions (Hamill, Palmer, & Emmerik, 2012). Limited movement variability reduces task complexity via decreasing the degrees of freedom used in the task, this in turn may reach a critical tissue threshold whereby injury occurs (Hamill et al., 2012). An injury may occur via the load being unable to be distributed across many structures or a shorter duration between loading events, increasing the likelihood of tissue maladaptation (James et al., 2000). Movement variability has been shown to discriminate runners with a history of low back pain from those without a history and current low back pain (Seay et al., 2011). It is unknown whether a player with a history of a groin pain, a groin injury risk factor (Engebretsen et al., 2010), will demonstrate an aberrant cut task strategy and/or lower movement variability during a cut task when compared to uninjured players.

Therefore, the aim of this exploratory study was to examine the cut task strategies and the magnitude of movement variability utilized during an unanticipated cut task by players with and without a history of groin pain. It was hypothesized that players with a history of groin pain would display altered lower limb kinematics, increased ground reaction forces and joint moments and lower movement variability during an unanticipated cut task, and decreased hip adduction muscular strength compared to players without a history of groin pain.

2. Method

2.1. Participants

Based on our previous research on hip joint kinematics during landing (Edwards et al., 2010), sample size was estimated for an error probability of 0.05 and statistical power of 80% using G*Power Software (Faul, Erdfelder, Lang, & Buchner, 2007). Utilizing the Bayesian approach only requires approximately one third of the sample size on those based on hypothesis testing (Batterham & Hopkins, 2006) and therefore a minimum of seven participants per group were deemed sufficient.

Seventeen first grade male Australian football players who had no current groin pain or history of major lower limb injuries that required surgery were recruited. Each participant was examined by a qualified physiotherapist to ensure they had no current groin pain and to classify the players as without (CONTROL; $n = 10$) or with (HISTORY; $n = 7$) a history of a groin pain. The physiotherapy examination included a squeeze test, rectus abdominis-related pain test, pubic symphysis related pain test, bent knee fall out and the flexion (hip flexibility), adduction, internal rotation test (hip flexibility) (Malliaras, Hogan, Nawrocki, Crossley, & Schache, 2009). Players completed a Groin Outcome Score (Engebretsen, Myklebust, Holme, Engebretsen, & Bahr, 2008) to assess groin function at the time of testing. A history of groin pain and severity was defined according to Werner et al. (2009) (Table 1). No details of the treatment previously undertaken in participants with a history of groin pain was recorded. It should be acknowledged that the minimum standards for reporting the injury description and examination of groin pain in athletes were not adhered to (Weir et al., 2015a) as these guidelines were developed after the completion of this study's data collection. Although an additional two participants diagnosed with current groin pain were recruited as participants within this study, their current pain affected their ability to perform the cut task, and therefore it was deemed inappropriate to continue data collection for this cohort of participants. Each participant gave written consent prior to participation and data collection with all methods approved by the institution's Human Research Ethics Committee.

2.2. Data collection

Participants performed hip abduction-adduction contractions that consisted of three sub-maximal warm-up efforts at $60^\circ \cdot s^{-1}$, $75^\circ \cdot s^{-1}$ and $90^\circ \cdot s^{-1}$ with 5 s between each repetition, immediately followed by five maximal isokinetic contractions with 1 min rest. Each lower limb was tested in a counterbalanced order using an isokinetic dynamometer (50 Hz, Humac Norm, CSMi Medical Solutions, Stoughton, USA) through a 0–30° hip abduction range of motion at $120^\circ \cdot s^{-1}$ in a side-lying position with a thigh and torso stabilization strap to minimize secondary movements.

Thirty-eight passive reflective markers were placed on each participant's lower limbs, pelvis, torso and own athletic shoes (Mann, Edwards, Drinkwater, & Bird, 2013). Participants performed a self-paced warm-up for 5–10 min on a cycle ergometer (Monark Model 828E, Sweden), and then completed 10 successful trials of a right ($n = 10$) and left ($n = 10$) unanticipated cut task in randomized order with an immobile defensive 'opponent'. During each trial, participant's three-dimensional lower limb and trunk motion kinematics were recorded (250 Hz) using an eight camera Qualisys Oqus 300+ camera system (Qualisys AB, Göteborg, Sweden). GRF data were recorded (1000 Hz) using a multichannel force platform (Type 9281CA, Kistler, Winterthur, Switzerland) embedded in the floor with a 20 mm athletic track surface and connected to a control unit with built-in charge amplifier (Type 5233A, Kistler, Winterthur, Switzerland). Kinematic and kinetic data were time synchronized and collected using Qualysis Track Manager (Version 2.6, Qualisys AB, Göteborg, Sweden).

2.3. Experiment task

The participant accelerated forward for 7 m towards the force platform, followed by a plant-and-cut maneuver at an angle between 35° and 55° originating at the center of the force plate with a skeleton placed 20 cm past the force platform to simulate a defensive opponent in a game environment. One of two colored light-emitting diodes, which indicated a right or left cut direction, was signaled manually ~2 m (individually adjusted) prior to the participant approaching the force platform. In a successful trial, the participants' foot wholly contacted the force platform during the plant-and-cut (opposite foot to the direction of the cut task (right foot for left cut task)), the participants cut direction matched the light-emitting diode, and the approach speed was between

Table 1
Description of the groin injury history of each participant within the HISTORY group.

Participant	Number of episodes	Location of pain	Groin outcome score at time of testing			Severity of injury[a]
			Symptoms	Pain	Quality of performance	
1	NA	Central	100%	100%	67%	Moderate
2	1	Central	92%	95%	73%	Moderate
3	1	Central	92%	80%	100%	Minimal
4	3	Left	44%	47%	40%	Severe
5	>5	Central	92%	97%	73%	Mild
6	>5	Right	76%	73%	93%	Mild
7	1	Central	96%	100%	100%	Mild

NA, not available.
[a] Define according to Werner (Werner et al., 2009).

4.5 m s^{-1} to 5.5 m s^{-1} (Speed Light; Swift Sports Equipment, Lismore, Australia). To ensure sufficient recovery and minimize effects of fatigue, participants were given 1 min rest between each trial, and a minimum of 5 min following every 20 trials.

2.4. Data reduction and analysis

As most HISTORY participants only reported previous central and/or right groin pain and were right lower limb dominant based on preferred kicking leg, only the left unanticipated cut task (right lower limb) was reported. Analysis of the kinematic and kinetic data was performed using Visual 3D software (Version 4, C-Motion, Maryland, USA). Raw kinematic coordinates, GRFs, free moments and center of pressure data initially filtered using a fourth-order zero-phase-shift Butterworth digital low pass filter (f_c = 18 Hz) before calculating individual joint kinematics and internal joint moments during a cut task. The raw kinetic data was used to calculate the individual GRFs and were normalized to a participant's body weight. Segment masses and inertial properties were defined using the procedures outlined in Mann et al. (2013). To maintain sign conventions through all segments, all right-handed Cartesian local coordinate systems were defined as x-axis medio-lateral, y-axis anterior-posterior and z-axis up direction with a *x,y,z* Cardan sequence utilized.

Using the 18 Hz filtered kinetic data, the weight acceptance phase was defined from initial contact (IC) when the vertical GRF exceeded 10 N, to the first local minimum (F_{WA}) after peak vertical GRF (F_V). The stance phase was defined from IC to toe-off when the vertical GRF fell below 10 N. The vertical GRF data were used to calculate temporal events (IC, F_V and F_{WA}).

2.5. Statistical analysis

As no previous research have investigated movement patterns in players with current or history of groin injury, no specific primary outcome variables were selected for this exploratory study. Outcome variables assessed during the stance phase of the cut task were the three-dimensional ankle, knee, hip, L5-S1 and T12-L1 joint kinematics, peak net internal ankle, knee and hip joint moments (normalized in accordance with the participant's body mass), F_V, F_{WA} and peak posterior GRF (F_{POST}).

Means and standard deviations were calculated for each kinetic and kinematic outcome variable during the cut task, and descriptive characteristics for the participants' with (HISTORY) and without (CONTROL) a history of groin pain during the weight acceptance phase of the cut task. After confirming normality with a Kolmogorov–Smirnov test with Lilliefors correction, the data was analyzed using a series of independent-samples *t*-tests to determine whether there were any between-group differences in the mean values of the outcome variables. Although multiple statistical tests were conducted, increasing the chance of inflation of the Type I error, no adjustment was deemed necessary (Hopkins, Marshall, Batterham, & Hanin, 2009) as these adjustments may increase the likelihood of Type II errors (Perneger, 1998), this present study was exploratory in nature and a Bayesian approach was used.

Clinical research should provide readers with information on the inferences about true (population) values they can probably expect from a given intervention. This is best expressed employing a Bayesian approach using precision of estimation (95% confidence limit) and magnitude-based inferences (effect sizes (Cohen, 1988)). These statistics were also employed to avoid the shortcomings of research based in null-hypothesis significance testing (Hopkins et al., 2009). Expressing results utilizing these methods indicates that whilst it is unknown the true effect of any intervention, they estimate that the true effect probability lies between the upper and lower limits. Then, by converting the absolute values of mean, lower and upper confidence limits to Cohen's *d* (effect sizes), it allows all variables to be compared on a common metric. Any effect with a substantial portion of the 95% confidence limit lying outside the range of −0.20 to 0.20 was considered to be unlikely to be trivial and therefore have a substantial likelihood (>75%) to be sufficiently large and consistent enough to be clinically meaningful. Such an approach makes questions of Type I and Type II errors moot since no inference is being made to statistical significance. Moderate or large effect sizes were defined as substantial changes as it enables more informative inferential assertions regarding magnitude (Hopkins et al., 2009).

Linear measure of variability was used to assess the magnitude of movement variability (Harbourne & Stergiou, 2009) between-group for each outcome variable during the cut task. This study employed linear methods as they investigate the magnitude of variability and can be utilized in non-cyclic movements (i.e. cut task), whereas non-linear methods explore the complexity of movement are complementary to linear tools (Harbourne & Stergiou, 2009) but can only been used in cyclic movements. To calculate movement variability, the data were analyzed using a series of repeated measures ANOVA using IBM SPSS statistical package (Version 17.0.1, SPSS Inc, Chicago, IL) to determine the within-subject error of each group to yield the coefficient of variation (CV) and expressed as a percent of the mean outcome variable. Then by calculating the likely limits for the ratio of the CV, the within-subject CV of the two groups were compared (Hopkins & Hewson, 2001). If the ratio of the CV of the two groups differed by a factor of greater than 1.15 or less than 0.85, it was deemed substantial (Hopkins & Hewson, 2001). All other procedures were conducted using a customized Excel spreadsheet (Hopkins, 2007).

3. Results

The HISTORY group were older, had a higher body mass, decreased right hip adductor-abductor muscular strength ratio,

lower right hip peak adductor muscular strength, and a decreased squeeze test at 30° when compared to the CONTROL group (Table 2). Interestingly, the HISTORY displayed significantly lower squeeze test values at 30° ($d = 1.06$; $P = 0.04$; 95%CL = 27) and 45° ($d = 1.13$; $P = 0.03$; 95%CL = 29) but not for the CONTROL group at 30° ($d = 0.10$; $P = 0.83$; 95%CL = 3) nor 45° ($d = 0.39$; $P = 0.39$; 95% CL = 9) when compared to 0°.

During the stance phase, the HISTORY group landed with a moderately less inversion angle and greater forefoot adduction angle at F_{WA} compared to the CONTROL group (Fig. 1, Table 3). In comparison to the CONTROL group, the HISTORY group displayed less knee flexion at IC (large), F_V (moderate) and F_{WA} (large), greater knee abduction at IC and F_{WA} (moderate), and greater knee internal rotation at IC (large), F_V (moderate) and F_{WA} (large) when compared to the CONTROL group. Less hip internal rotation was displayed by the HISTORY at IC (moderate), F_V and F_{WA} (large) when compared to the CONTROL group (Fig. 2). The L5-S1 joint angle showed moderately smaller effect for flexion at F_{WA} and greater effect for right rotation at IC when comparing the HISTORY to the CONTROL group. While the T12-L1 joint consistently reported effects for greater flexion at F_V and F_{WA}, greater left lateral flexion at IC, F_V and F_{WA} (moderate), and greater right rotation at IC (large), F_V and F_{WA} (moderate) for HISTORY compared to the CONTROL group.

GRFs were consistently higher in the HISTORY group compared to the CONTROL group for F_V and F_{POST}, (moderate) and F_{WA} (large; Table 4). There were no substantial between-group differences for any peak joint moments.

Throughout the stance phase, the HISTORY group displayed lower movement variability than the CONTROL group for all of ankle and knee joint angle variables, except for forefoot adduction-abduction at F_{WA} and knee adduction-abduction at F_V, and most L5-S1 and T12-L1 joint kinematics (Table 3).

The HISTORY group displayed substantially lower movement variability for GRFs, and most peak ankle, knee and hip joint moments variables during the stance phase, except for peak ankle inversion, knee abduction and hip adduction joint moments, which displayed substantially higher movement variability when compared to the CONTROL group (Table 4).

4. Discussion

Different cut strategies associated with lower limb injury (McLean et al., 2004; Scase et al., 2006) were observed in this study in participants with a history of groin pain compared to participants without a history of groin pain. Players with a history of groin pain also displayed lower ankle, knee and T12-L1 joints kinematic movement variability but surprisingly higher hip and L5-S1 joints kinematic movement variability compared to those without a history. This suggests substantial between-group differences for most kinematic and kinetic variables, and the contributions of these joints to the way an unanticipated cut task was executed. It should be acknowledged that this study's findings may have been confounded due to the retrospective nature of this study and its lack of adherence to the new minimum standards for reporting groin pain in athletes that were developed after the completion of this study's data collection. That is, the disparity in the diagnostic terminology of groin pain and poor level of agreement between experts on a patient case (Weir, Hölmich, Schache, Delahunt, & de Vos, 2015b), may have led to some of the history participants to be incorrectly diagnosed with groin pain, and the clinical entity of the history of groin pain was unknown.

Landing with a greater plantar-flexion angle has been associated with improved force dissipation (Self & Paine, 2001) via increasing range of ankle joint motion available to decelerate the body. Although the HISTORY group could have utilized this strategy to decelerate the body, they displayed a similar ankle joint range of motion throughout the stance phase compared to the CONTROL group. They also utilized an invariant ankle and knee joint landing strategy, via decreased ankle and knee joint kinematic movement variability. This may have a limited their force management strategy at the ankle joint that prevented them from using an ankle control strategy (McNitt-Gray, Requejo, & Flashner, 2006) of the CONTROL group, and perhaps increased transfer of force to the groin region. Although these considerable joint loads sustained during the cut task can be modified by either reducing approach speed or decreasing cut angle (Besier, Lloyd, Cochrane, & Ackland, 2001b), these two factors were controlled within the study task requirements.

Increased knee abduction and internal rotation, decreased knee flexion, and increased T12-L1 flexion, left lateral flexion and right rotation angles were displayed by the HISTORY group compared to the CONTROL group. Although these knee (McLean et al., 2004) and trunk (Dempsey et al., 2007; Jamison, Pan, & Chaudhari, 2012) variables are predisposing risk factor towards anterior cruciate

Table 2
Characteristics of players with and without a history of groin pain.

Variable	Control	History	d_a	CL	P
Age (years)	20.8 ± 1.1	22.5 ± 3.8	0.66[a]	2.7	0.19
Height (cm)	181.2 ± 4.3	181.7 ± 8.2	0.09	6.5	0.86
Body Mass (kg)	79.5 ± 5.3	84.5 ± 13.2	0.54[a]	9.8	0.29
Right Leg Flexibility (cm)	16.4 ± 1.8	14.6 ± 3.9	0.47	4.1	0.36
Left Leg Flexibility (cm)	15.8 ± 1.8	13.9 ± 4.1	0.44	4.3	0.38
Right					
Hip adductor/abductor ratio (%)	106 ± 39	79 ± 21	0.78[a]	47	0.13
Hip abductor peak ($Nm \cdot kg^{-1}$)	1.04 ± 0.29	1.11 ± 0.45	0.19	0.51	0.71
Hip adductor peak ($Nm \cdot kg^{-1}$)	1.11 ± 0.47	0.86 ± 0.36	0.59[a]	0.55	0.26
Left					
Hip adductor/abductor ratio (%)	93 ± 33	82 ± 25	0.36	33	0.50
Hip abductor peak ($Nm \cdot kg^{-1}$)	1.12 ± 0.29	1.18 ± 0.47	0.15	0.57	0.78
Hip adductor peak ($Nm \cdot kg^{-1}$)	1.07 ± 0.50	0.93 ± 0.38	0.32	0.57	0.55
Squeeze test (mmHg)					
0° hip flexion	109 ± 24	119 ± 23	0.43	44	0.40
30° hip flexion	106 ± 27	91 ± 23	0.60[a]	16	0.24
45° hip flexion	100 ± 22	89 ± 20	0.48	40	0.35
Approach speed ($m \cdot s^{-1}$)	5.0 ± 0.2	5.1 ± 0.2	0.47	0.2	0.36

95% confidence limit (CL) defines the range representing the uncertainty in the true value of the (unknown) population mean.
d_a effect size.
[a] Indicates moderate between-group condition difference in the effect size (value 0.50–0.79). [b] Indicates large between-group condition difference in the effect size (value ≥ 0.80). [c] Indicates a significant between-group difference, $P < 0.05$.

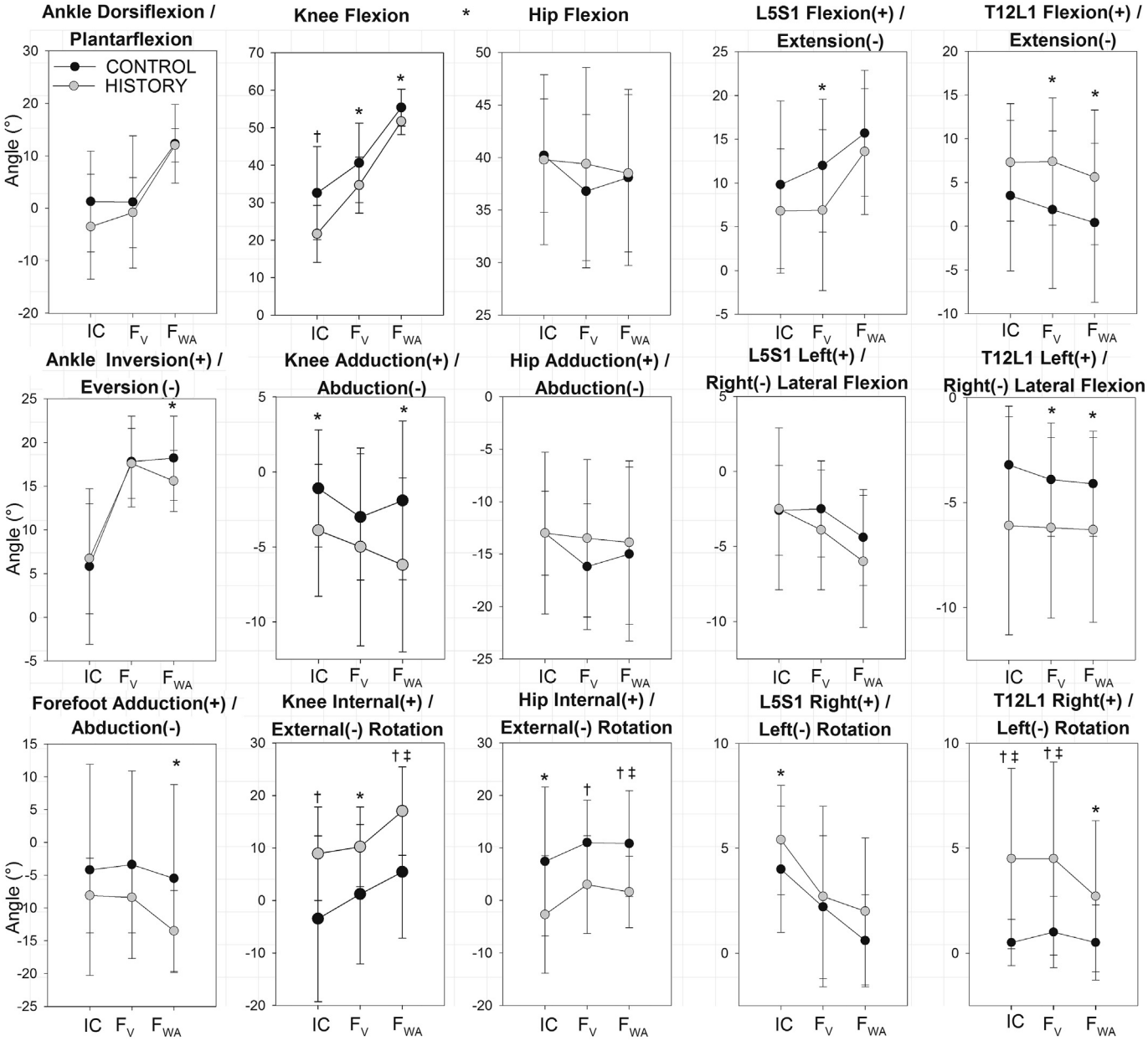

Fig. 1. Joint angle (°) values displayed during the weight acceptance phase of an unanticipated cut task for players with (HISTORY) and without (CONTROL) a history of groin pain. Initial foot-ground contact (IC), peak vertical ground reaction force (F_V), and first local minimum of the vertical ground reaction force after F_V (F_{WA}), and 95% confidence limit (CL) defines the range representing the uncertainty in the true value of the (unknown) population mean. *indicates moderate between-group condition difference in the effect size (value 0.50–0.79). † indicates large between-group condition difference in the effect size (value ≥ 0.80). ‡ indicates a significant between-group condition difference, $P < 0.05$.

ligament injury, this study's results do not suggest that players with a history of a groin injury are at an increased risk this knee injury. Instead it is postulated that players who display different cut strategies are at an increased risk of a lower limb and/or lumbopelvic injury, the injury(s) that they are at an increased risk of sustaining will depend on other specific risk factors associated with each injury.

Groin injury history is often associated with decreased hip internal rotation (Tak et al., 2015; Verrall et al., 2005; Verrall et al., 2007) with lower range of hip motion associated greater reported hip and groin-related problems (Tak et al., 2015). This may explain the neutral hip position maintained by the HISTORY group compared to the larger hip internal rotation displayed by the CONTROL group. This decreased movement has also been associated with increased demands on the pelvis and therefore the pubic symphysis, subsequently increasing the risk of groin injury (Verrall et al., 2005). The neutral position within the HISTORY group may also have been due to the decreased ability of the hip adductor muscle group to control the hip while executing the cut task. As the hip is flexed ~40° during the cut task, the significantly weaker squeeze test results at 30° and 45° than 0° displayed by the HISTORY group may have contributed to their compromised ability to control lower limb function and segmental alignment during the cut task.

Due to their key role in the lumbopelvic and hip stability and control over lower limb segmental alignment (Hrysomallis, 2009), hip adductor function may have contributed to the increased hip and L5-S1 joint kinematic movement variability but hip abduction

Table 3
Joint angles and between-group differences in movement variability values displayed during the weight acceptance phase of an unanticipated cut task for players with and without a history of groin pain.

Joint variable	At the time of IC							At the time of the peak F_V							At the time of the F_{WA}						
	Control		History		d^a	CL	P	Control		History		d^a	CL	P	Control		History		d^a	CL	P
	Error	CV%	Error	CV%				Error	CV%	Error	CV%				Error	CV%	Error	CV%			
Ankle																					
Dorsi-plantarflexion	43.9	10.9	23.7	8.9	0.49	10.2	0.34	67.5	14.9	27.2	9.8	0.19	11.2	0.71	35.4	9.0	11.7	5.4	0.06	6.5	0.91
Inversion-eversion	74.8	288.2	25.2	78.3	0.12	8.4	0.82	67.1	1732.0	25.9	137.2	0.03	5.0	0.95	54.1	303	81.2	87.2	0.59[a]	4.5	0.25
Forefoot adduction-abduction	23.1	37.4	15.0	21.9	0.30	13.6	0.56	60.6	34.5	51.7	26.5	0.43	12.2	0.40	28.7	23.7	42.3	23.1	0.66[a]	12.4	0.19
Knee																					
Flexion-extension	189.5	42.3	30.5	25.8	0.92[b]	11.3	0.06	61.2	19.2	31.4	14.5	0.61[a]	10.0	0.22	33.9	10.4	12.0	6.7	0.78[a]	4.6	0.12
Adduction-abduction	17.7	372.8	13.7	93.1	0.67[a]	4.3	0.18	14.6	129.0	41.7	119.2	0.38	5.6	0.46	12.3	186.9	30.0	110.4	0.74[a]	5.8	0.14
Internal-external rotation	146.2	360.9	86.3	105.1	0.86[b]	14.2	0.08	95.2	802.5	89.4	80.7	0.75[a]	11.9	0.13	67.3	223.4	116.0	63.6	0.95[b]	11.7	0.05[c]
Hip																					
Flexion-extension	64.0	19.5	51.8	18.0	0.06	7.0	0.91	66.2	21.8	191.5	35.5	0.32	8.5	0.54	81.0	23.5	37.5	16	0.06	8.4	0.91
Adduction-abduction	18.6	33.0	25.7	40.7	0.01	6.0	0.99	26.9	32.6	48.3	50.4	0.41	7.0	0.42	26.8	34.6	34.2	42.5	0.14	8.5	0.78
Internal-external rotation	106.6	137.7	132.5	510.3	0.74[a]	13.7	0.14	63.4	68.4	111.2	341.9	0.87[b]	9.0	0.08	36.6	53.7	52.1	522.9	0.94[b]	9.4	0.05[c]
L5-S1																					
Flexion-extension	132.6	123.4	158.5	143.2	0.36	9.1	0.49	31.0	58.8	323.5	211.6	0.61[a]	8.7	0.23	29.5	42.8	62.1	55.0	0.31	7.5	0.55
Left-right lateral flexion	12.1	114.1	17.7	127.7	0.02	4.4	0.97	9.6	115.8	23.1	121.8	0.40	3.7	0.43	4.7	45.9	20.7	−70.1	0.44	3.9	0.39
Right-left rotation	19.0	126.7	33.9	138.1	0.50[a]	3.0	0.33	13.1	156.9	79.8	466.8	0.14	4.0	0.79	19.0	502.7	26.3	363.4	0.48	2.9	0.34
T12-L1																					
Flexion-extension	170.2	162.2	69.8	106.0	0.47	8.3	0.35	31.0	107.3	15.4	61.6	0.64[a]	8.8	0.20	25.5	128.4	29.7	103.6	0.57[a]	9.5	0.26
Left-right lateral flexion	4.6	71.6	6.8	45.3	0.69[a]	4.2	0.17	5.1	63.1	7.3	45.9	0.63[a]	3.6	0.21	3.4	50.1	8.2	49.3	0.63[a]	3.6	0.21
Right-left rotation	7.9	589.0	16.0	100.3	1.17[b]	3.0	0.01[c]	7.2	221.8	150.3	327.4	0.99[b]	3.4	0.04[c]	5.2	267.0	6.7	111.0	0.78[a]	2.8	0.12

95% confidence limit (CL) defines the range representing the uncertainty in the true value of the (unknown) population mean, within-subject (error) variance (Error), within subject coefficient of variation percentage (CV%), d_a effect size.

[a] Indicates moderate between-group condition difference in the effect size (value 0.50–0.79).

[b] Indicates large between-group condition difference in the effect size (value $\geq$ 0.80).

[c] Indicates a significant between-group difference, $P < 0.05$.

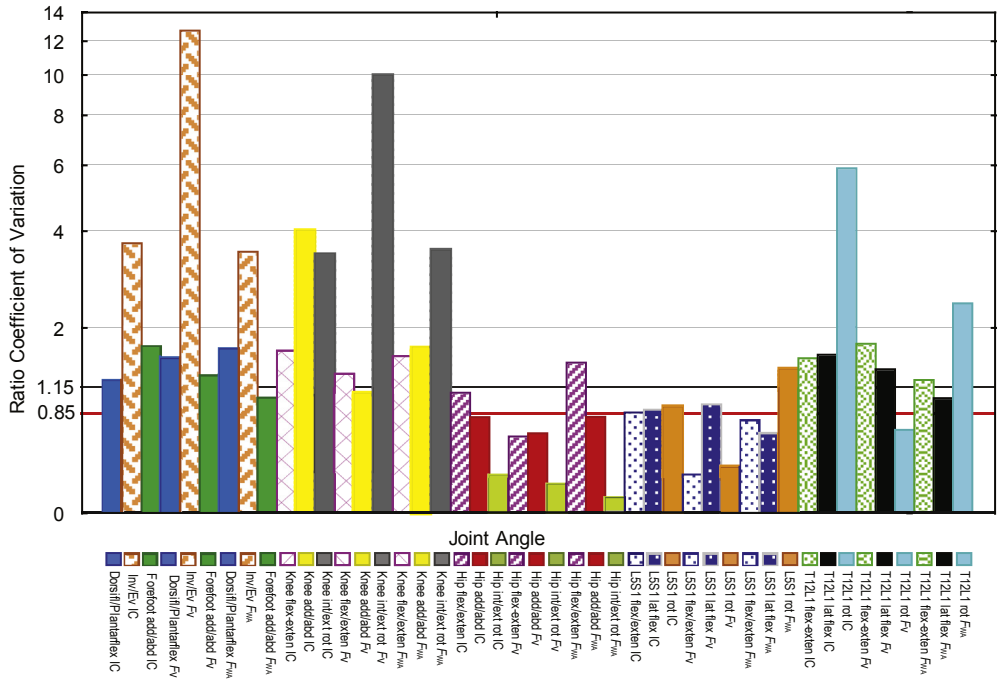

Fig. 2. Within-subject coefficient of variation ratio for joint angles between players with and without a history of groin pain during the weight acceptance phase of an unanticipated cut task. The HISTORY group displaying lower (>1.15) or higher (<0.85) variability.

Table 4
Peak ground reaction force (normalized to body weight) and net internal joint moment ($N \cdot kg^{-1}$) and between-group differences in movement variability values sustained during the weight acceptance phase of an unanticipated cut task for players with and without a history of groin pain.

Variable	Control	History	d_a	CL	*P*	Control		History		CV ratio
						Error	CV%	Error	CV%	
Force (BW)										
F_V	2.8 ± 0.8	3.1 ± 0.3	0.52[a]	0.68	0.30	0.30	18.6	0.20	14.2	1.31[c]
F_{WA}	2.0 ± 0.2	2.2 ± 0.2	0.82[b]	0.23	0.10	0.05	10.3	0.02	6.8	1.51[c]
F_{POST}	0.7 ± 0.2	0.8 ± 0.1	0.50[a]	0.16	0.33	0.02	16.1	0.01	−8.6	1.88[c]
Ankle joint moment ($Nm \cdot kg^{-1}$)										
Plantarflexion	1.67 ± 0.93	1.57 ± 0.74	0.13	0.90	0.81	0.27	32	0.12	22	1.43[d]
Forefoot abduction	0.39 ± 0.26	0.32 ± 0.16	0.30	0.24	0.56	0.08	73	0.04	61	1.20[d]
Inversion	0.41 ± 0.21	0.47 ± 0.22	0.25	0.22	0.63	0.03	39	0.14	81	0.48[c]
Knee joint moment ($Nm \cdot kg^{-1}$)										
Extension	3.51 ± 0.52	3.72 ± 0.39	0.44	0.50	0.39	0.34	17	0.29	14	1.16[d]
Adduction	0.94 ± 0.30	1.05 ± 0.67	0.24	0.51	0.65	0.14	41	0.21	43	0.96
Abduction	0.42 ± 0.35	0.59 ± 0.42	0.45	0.40	0.38	0.13	85	0.35	104	0.82[c]
External rotation	0.70 ± 0.27	0.76 ± 0.45	0.17	0.37	0.74	0.07	39	0.08	38	1.02
Hip joint moment ($Nm \cdot kg^{-1}$)										
Flexion	4.67 ± 1.78	5.30 ± 1.78	0.32	2.22	0.55	4.18	43	3.52	36	1.21[d]
Adduction	1.28 ± 0.52	1.68 ± 1.56	0.40	1.13	0.46	0.69	63	1.65	75	0.84[c]
Abduction	1.01 ± 1.01	1.10 ± 0.83	0.10	1.05	0.85	1.15	106	0.03	49	2.17[d]
External rotation	1.99 ± 0.73	2.58 ± 2.60	0.36	1.84	0.50	0.66	40	1.06	46	0.87

Peak vertical ground reaction force (F_V), first local minimum of the vertical ground reaction force after F_V (F_{WA}), peak posterior ground reaction force (F_{POST}), 95% confidence limit (CL) defines the range representing the uncertainty in the true value of the (unknown) population mean, within-subject (error) variance (Error), within subject coefficient of variation percentage (CV%), d_a effect size.
[e]Indicates a significant between-group difference, $P < 0.05$.
[a] Indicates moderate between-group difference in the effect size (value 0.50–0.79).
[b] Indicates large between-group difference in the effect size (value > 0.80).
[c] Indicates substantial difference in the within-subject CV with a *CV ratio > 1.15* and the HISTORY group displaying lower variability.
[d] Indicates substantial difference in the within-subject CV with a *CV ratio < 0.85* and the CONTROL group displaying lower variability.

function did not contribute. It remains unknown whether the weaker right hip adductor muscle strength displayed by the HISTORY group compared to the CONTROL group may have preceded the groin injury or arose from the injury. Increased lumbopelvic kinematic movement variability cannot be explained by the hip abductors, which provides lumbopelvic and hip stability (Hrysomallis, 2009) and eccentrically controls hip adduction and internal rotation during weight bearing, due to the lack of between-group differences in hip abduction muscular strength. Nevertheless, as hip isokinetic muscular strength testing was performed concentrically, it remains unknown if there were any between-group differences in the eccentric muscular strength of hip abductors.

As movement variability needs to occur somewhere along the

kinetic chain (Winter 1984), perhaps due to the coupled invariance of the ankle and knee segments, the HISTORY group utilized increased lumbopelvic kinematic movement variability (hip and L5-S1) compared to the CONTROL group. Despite this hip and L5-S1 joint adaptation, it occurred at the critical times of peak loading and resulted in decreased movement variability of the magnitude of the peak hip flexion and abduction joint moments. This suggests that when stability and control are critical during landing, the lumbopelvic region in the HISTORY group may be functioning with a limited force dissipation strategy that increases the likelihood of tissue maladaptation. This difference in lower limb movement variability supports previous research in players with chronic ankle instability who displayed increased movement variability at the previously injured joint (ankle) but decreased variability in other lower limb joints (knee and hip joints) during stop-jump landing (Brown et al., 2012; Brown, Padua, Marshall, & Guskiewicz, 2009).

The HISTORY group appeared to exhibit an adaptation by using their thoracic segment to maintain movement control equilibrium and return the center of body mass over the base of support, in this case the stance limb. This compensation at the T12-L1 joint showed increased flexion, left lateral flexion and right rotation amongst the HISTORY compared to the CONTROL. These differences, particularly within the T12-L1 joint, support the concept of a distinct landing strategy used to maintain the body's center of support. As the body's largest mass segment, the thorax segment only needs to move slightly to create a large clockwise torque through right laterally flexing the T12-L1 joint to counteract the large anticlockwise torque imposed by the left side of the pelvis dropping due to increased lumbopelvic movement. Therefore, only small movement adjustment via decreased movement variability of the T12-L1 joint is needed to reposition the center of gravity over the stance limb to induce an effective clockwise torque to maintain movement stability and the center of gravity within the base of support.

Interestingly, the higher peak GRF, decreased hip adduction muscular strength combined with the different kinematics during the cut task of the HISTORY group compared to the CONTROL group lead to no substantial between-group differences in any peak joint moments. This supports previous research that peak GRFs should not be used to infer peak joint moments (Edwards et al., 2012), and that peak joint moments during a cut task do not alter with increased trunk range of motion (Edwards, Austin, & Bird, in press) but are in disagreement with increased knee joint moment observed (Dempsey et al., 2007; Jamison et al., 2012).

Although the standard deviation is a valid (Altman & Bland, 2005) linear measure of variability, the CV was chosen to estimate changes in the magnitude of movement variability (Harbourne & Stergiou, 2009) as it is a dimensionless measure that enables comparisons between different populations. Nevertheless, the CV becomes much more sensitive to change when the mean approaches zero (Abdi, 2010; Brown et al., 2012). Forefoot adduction-abduction at F_V for example, the CONTROL group displayed a mean joint angle of -0.5° with a within-subject (error) variance of only 67.1 resulting in a CV of 1732%. Although this highlights that the magnitude of the CV may be not is not a true reflection when the mean approaches zero, most of the variables are in agreement when comparing the differences of the within-subject (error) variance between-groups with the CV ratio.

5. Conclusions

Cause or effect has not been established in this exploratory cross-sectional study and this study did not adhere to the new minimum standards for reporting groin pain in athletes. It is postulated that the poor lower limb segmental alignment and these three invariant systems (ankle and knee joints, and T12-L1 joint) do not adapt their movement kinematics of the task to the environmental constraints, while being connected by a segment (hip and L5-S1 joints) that displays increased lumbopelvic movement. Weak hip adductor muscular strength may have served as a precursor for this distinct cut task strategy. This provides evidence to suggest that lower limb segmental alignment and control during an agility task should also be included within the management. This management should be ensured further than the resolution of pain (Seay et al., 2011) to reduce the loss of complexity of the movement task that may contribute to the high reoccurrence rates of groin injuries.

Funding

The authors declare that that no external financial support was received.

Conflict of interest

The authors have no conflicts of interest that are directly related to the context of this article.

Ethical approval

Charles Sturt University Human Research Ethics Committee approved this study. All participants received verbal and written information about the study and gave written informed consent prior to data collection.

Acknowledgements

The authors acknowledge Mr Luke Howard (Bathurst Physiotherapy, Australia) for involvement within this studies participant diagnosis and physiotherapist consultations, and Dr Eric Drinkwater for his statistical guidance. Prof Cook was supported by the Australian Centre for Research into Sports Injury and its Prevention, which is one of the International Research Centres for Prevention of Injury and Protection of Athlete Health supported by the International Olympic Committee (IOC). Prof Cook is supported by a NHMRC practitioner fellowship (1058493).

References

Abdi, A. (2010). Coefficient of variation. In N. Salkind (Ed.), *Encyclopedia of research design* (pp. 169–171). Thousand Oaks, CA: SAGE Publications Inc.

Altman, D. G., & Bland, J. M. (2005). Standard deviations and standard errors. *British Medical Journal, 331*, 903.

Batterham, A. M., & Hopkins, W. G. (2006). Making meaningful inferences about magnitudes. *International Journal of Sports Physiology & Performance, 1*, 50–57.

Besier, T. F., Lloyd, D. G., Ackland, T. R., & Cochrane, J. L. (2001a). Anticipatory effects on knee joint loading during running and cutting maneuvers. *Medicine & Science in Sports & Exercise, 33*, 1176–1181.

Besier, T. F., Lloyd, D. G., Cochrane, J. L., & Ackland, T. R. (2001b). External loading of the knee joint during running and cutting maneuvers. *Medicine & Science in Sports & Exercise, 33*, 1168–1175.

Brown, C., Bowser, B., & Simpson, K. J. (2012). Movement variability during single leg jump landings in individuals with and without chronic ankle instability. *Clinical Biomechanics, 27*, 52–63.

Brown, C. N., Padua, D. A., Marshall, S. W., & Guskiewicz, K. M. (2009). Variability of motion in individuals with mechanical or functional ankle instability during a stop jump maneuver. *Clinical Biomechanics, 24*, 762–768.

Chaudhari, A. M. W., Jamison, S. T., McNally, M. P., Pan, X., & Schmitt, L. C. (2014). Hip adductor activations during run-to-cut manoeuvres in compression shorts: Implications for return to sport after groin injury. *Journal of Sports Sciences, 32*, 1333–1340.

Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd ed.). Hillsdale: N.J: L. Erlbaum Associates.

Dempsey, A. R., Lloyd, D. G., Elliott, B. C., Steele, J. R., Munro, B. J., & Russo, K. A. (2007). The effect of technique change on knee loads during sidestep cutting. *Medicine & Science in Sports & Exercise, 39*, 1765–1773.

Drew, M. K., Osmotherly, P. G., & Chiarelli, P. E. (2014). Imaging and clinical tests for

the diagnosis of long-standing groin pain in athletes. A systematic review. *Physical Therapy in Sport, 15*, 124–129.
Edwards, S., Austin, A. P., & Bird, S. P. (2016). The role of the trunk control in athletic performance of a reactive change-of-direction task. *Journal of Strength & Conditioning Research*. http://dx.doi.org/10.1519/JSC.0000000000001488 (in press).
Edwards, S., Steele, J. R., McGhee, D. E., Beattie, S., Purdam, C., & Cook, J. L. (2010). Landing strategies of athletes with an asymptomatic patellar tendon abnormality. *Medicine & Science in Sports & Exercise, 42*, 2072–2080.
Edwards, S., Steele, J. R., McGhee, D. E., Cook, J. L., Purdam, C., & Munro, B. J. (2012). Characterising patellar tendon loads during dynamic landings. *Scandinavian Journal of Medicine & Science in Sports, 22*, 2–11.
Engebretsen, A. H., Myklebust, G., Holme, I., Engebretsen, L., & Bahr, R. (2008). Prevention of injuries among male soccer players: A prospective, randomized intervention study targeting players with previous injuries or reduced function. *American Journal of Sports Medicine, 36*, 1052–1060.
Engebretsen, A. H., Myklebust, G., Holme, I., Engebretsen, L., & Bahr, R. (2010). Intrinsic risk factors for groin injuries among male soccer players: A prospective cohort study. *American Journal of Sports Medicine, 38*, 2051–2057.
Faul, F., Erdfelder, E., Lang, A.-G., & Buchner, A. (2007). G*Power 3: A flexible statistical power analysis program for the social, behavioral, and biomedical sciences. *Behavior Research Methods, 39*, 175–191.
Gabbe, B. J., Finch, C. F., Wajswelner, H., & Bennell, K. L. (2004). Predictors of lower extremity injuries at the community level of Australian football. *Clinical Journal of Sport Medicine, 14*, 56–63.
Hamill, J., Palmer, C., & Van Emmerik, R. E. (2012). Coordinative variability and overuse injury. *Sports Medicine, Arthroscopy, Rehabilitation, Therapy & Technology, 4*, 45.
Harbourne, R. T., & Stergiou, N. (2009). Movement variability and the use of nonlinear tools: Principles to guide physical therapist practice. *Physical Therapy, 89*, 267–282.
Hopkins, W. G. (2007). A spreadsheet for deriving a confidence interval, mechanistic inference and clinical inference from a p value. *Sportscience, 11*, 16–20.
Hopkins, W. G., & Hewson, D. J. (2001). Variability of competitive performance of distance runners. *Medicine & Science in Sports & Exercise, 33*, 1588–1592.
Hopkins, W. G., Marshall, S. W., Batterham, A. M., & Hanin, J. (2009). Progressive statistics for studies in sports medicine and exercise science. *Medicine & Science in Sports & Exercise, 41*, 3–13.
Hrysomallis, C. (2009). Hip adductors' strength, flexibility, and injury risk. *Journal of Strength & Conditioning Research, 23*, 1514–1517.
James, C. R., Dufek, J. S., & Bates, B. T. (2000). Effects of injury proneness and task difficulty on joint kinetic variability. *Medicine & Science in Sports & Exercise, 32*, 1833–1844.
Jamison, S. T., Pan, X., & Chaudhari, A. M. W. (2012). Knee moments during run-to-cut maneuvers are associated with lateral trunk positioning. *Journal of Biomechanics, 45*, 1881–1885.
Malliaras, P., Hogan, A., Nawrocki, A., Crossley, K., & Schache, A. (2009). Hip flexibility and strength measures: Reliability and association with athletic groin pain. *British Journal of Sports Medicine, 43*, 739–744.
Mann, K. J., Edwards, S., Drinkwater, E. J., & Bird, S. P. (2013). A lower limb assessment tool for athletes at risk of developing patellar tendinopathy. *Medicine & Science in Sports & Exercise, 45*, 527–533.
McLean, S. G., Lipfert, S. W., & van den Bogert, A. J. (2004). Effect of gender and defensive opponent on the biomechanics of sidestep cutting. *Medicine & Science in Sports & Exercise, 36*, 1008–1016.
McNitt-Gray, J. L., Requejo, P. S., & Flashner, H. (2006). Multijoint control strategies transfer between tasks. *Biological Cybernetics, 94*, 501–510.
Orchard, J., Seward, H., & Orchard, J. (2014). *2014 Australian football league injury report*. Melbourne, Australia.
Perneger, T. V. (1998). What's wrong with Bonferroni adjustments. *British Medical Journal, 316*, 1236–1238.
Scase, E., Cook, J., Makdissi, M., Gabbe, B., & Shuck, L. (2006). Teaching landing skills in elite junior Australian football: Evaluation of an injury prevention strategy. *British Journal of Sports Medicine, 40*, 834–838.
Seay, J. F., Van Emmerik, R. E. A., & Hamill, J. (2011). Low back pain status affects pelvis-trunk coordination and variability during walking and running. *Clinical Biomechanics, 26*, 572–578.
Self, B. P., & Paine, D. (2001). Ankle biomechanics during four landing techniques. *Medicine & Science in Sports & Exercise, 33*, 1338–1344.
Tak, I., Glasgow, P., Langhout, R., Weir, A., Kerkhoffs, G., & Agricola, R. (2015). Hip range of motion is lower in professional soccer players with hip and groin symptoms or previous injuries, independent of cam deformities. *American Journal of Sports Medicine, 44*, 682–688.
Thorborg, K., Rathleff, M. S., Petersen, P., Branci, S., & Hölmich, P. (2015). Prevalence and severity of hip and groin pain in sub-elite male football: A cross-sectional cohort study of 695 players. *Scandinavian Journal of Medicine & Science in Sports*. http://dx.doi.org/10.1111/sms.12623 (in press).
Verrall, G. M., Hamilton, I. A., Slavotinek, J. P., Oakeshott, R. D., Spriggins, A. J., Barnes, P. G., et al. (2005). Hip joint range of motion reduction in sports-related chronic groin injury diagnosed as pubic bone stress injury. *Journal of Science & Medicine in Sport, 8*, 77–84.
Verrall, G. M., Slavotinek, J. P., Barnes, P. G., Esterman, A., Oakeshott, R. D., & Spriggins, A. J. (2007). Hip joint range of motion restriction precedes athletic chronic groin injury. *Journal of Science & Medicine in Sport, 10*, 463–466.
Weir, A., Brukner, P., Delahunt, E., Ekstrand, J., Griffin, D., Khan, K. M., et al. (2015a). Doha agreement meeting on terminology and definitions in groin pain in athletes. *British Journal of Sports Medicine, 49*, 768–774.
Weir, A., Hölmich, P., Schache, A. G., Delahunt, E., & de Vos, R.-J. (2015b). Terminology and definitions on groin pain in athletes: Building agreement using a short Delphi method. *British Journal of Sports Medicine, 49*, 825–827.
Werner, J., Hagglund, M., Walden, M., & Ekstrand, J. (2009). UEFA injury study: A prospective study of hip and groin injuries in professional football over seven consecutive seasons. *British Journal of Sports Medicine, 43*, 1036–1040.
Winter, D. A. (1984). Kinematic and kinetic patterns in human gait: Variability and compensating effects. *Human Movement Science, 3*, 51–76.